

DAIYANG He

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 B.S. in Industrial Engineering  Sichuan University  2005-06-11  Chengdu, China 
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My research interest lies in **mechanism-informed AI for clinical insights**. My undergraduate work spans medical image analysis and hemodynamic simulation. The combination of data-driven and physics-based perspectives is my distinctive edge in developing physics-informed, clinically grounded AI systems. I am highly self-motivated, with strong analytical skills and a persistent drive for innovation supported by scientific research experience. Please see my homepage above for more my information.

Education

Present	SCUPI, Sichuan University
2023	B.S. (GPA: 3.85, 8/73) in Industrial Engineering

Competences & Languages

Programming	Python, C/C++, MATLAB, SQL, $\LaTeX$
Scientific Knowledge	Medical Image Analysis, Machine Learning, Harness Engineering, CFD, Probability & Statistics, Mathematical Analysis
Software	Mimics, 3D Slicer, ANSYS Fluent, ICEM CFD, ParaView, TecPlot
 Languages	English (fluent reading, writing, speaking)

Research Papers

- › S. Cheng, L. Liu, Y. Qiu, Y. Diao, **Daiyang He**, F. Yan, C. Wang, K. Li. *Hypoxia promotes blood blister-like aneurysms development: A CFD study*. *Journal of Biomechanics*, 2025.
- › S. Cheng, Y. Qiu, **Daiyang He**, M. L. Wang, C. Weng, D. Yuan, K. Li. *Helical vortex arising from aberrant right subclavian artery induces occurrence of dissection tears*. *Physics of Fluids*, 2025.
- › C. Weng*, **Daiyang He***, S. Cheng, D. Yuan, T. Wang, W. Zeng, Y. Luo, J. Wang, K. Li, Y. Qiu. *Noise-Pressure Constrained Liutex Method for Robust Vortex Identification in 4D Flow MRI of Abdominal Aortic Aneurysms*. *Computer Methods and Programs in Biomedicine*, Major Revision, 2026.
- › Y. Qiu*, **Daiyang He***, Z. Jiang, S. Cheng, Y. Tai, K. Li, H. Wu. *Open-source and fully automated multi-organ CT analysis for non-invasive portal hypertension evaluation*. *npj Digital Medicine*, Under Review, 2026.

Academic Competitions and Achievements

- › 2024: **Third Prize**, Southwest Division of the China Undergraduate Physics Tournament (CUPT)
- › 2024: **S Award**, American Collegiate Mathematical Contest in Modeling
- › 2025: **Third Prize**, National Collegiate Mathematical Contest in Modeling
- › 2025: **Undergraduate Training Program (National Level)**, "ShadowSift: An Efficient Real-Time Detection System for Low-Quality Deepfake Videos."
- › 2024-2026: **Excellent Student, Outstanding Students Leader, Comprehensive Scholarship**, Sichuan University

Work Experience

Present	Research Assistant, SCUPI BioFluid Lab
2023	› Conduct CT/MRI image analysis and CFD simulation. › Develop the web-based automatic portal hypertension assessment pipeline ALPHA (http://scupi.alpha.hxmedbot.com:10910/). › Explore physics-informed machine learning modeling for CFD surrogate.